

LESSON 9-3 • SESSION 1 • PRACTICE • FORM A

Equations from Tables

Learning target – I can write an equation that matches a table of values. • 6.EE.C.9

REMEMBER

- A **constant rate** means $y \div x$ is the same in every row. Then $y = (\text{rate}) \times x$.
- If $y - x$ is the same in every row, $y = x + (\text{that number})$.
- If neither works, see how much y grows each step: x : 1, 2, 3 and y : 5, 7, 9 gives $y = 2x + 3$.

WORKED EXAMPLE

Write an equation for the pairs (1, 6), (2, 12), (5, 30).

Step 1 $6 \div 1 = 6$, $12 \div 2 = 6$, $30 \div 5 = 6$.**Step 2** The rate is 6 every time.**Step 3** Check: $6 \times 5 = 30$.**Answer** $y = 6x$ **WORDS** equation (*ecuación*) • constant (*constante*) • rate (*tasa*)

PRACTICE

show your work • label units

1. Write an equation for each table.

a. x : 1, 2, 3 y : 5, 10, 15 $y =$ _____

b. x : 1, 2, 3 y : 12, 24, 36 $y =$ _____

c. x : 1, 2, 3 y : 4, 5, 6 $y =$ _____

d. x : 2, 4, 6 y : 18, 36, 54 $y =$ _____

- 2.
- Skate Park Passes.**
- Each pass costs the same. Write an equation for the cost
- y
- of
- x
- passes. Then find the cost of 7 passes.

Hint Divide a cost by its number of passes.

Passes (x)	1	2	3	5
Cost in \$ (y)	8	16	24	40

Equation; cost _____

- 3.
- Tile Pattern.**
- A floor uses the same number of tiles in every row. A table shows
- x
- : 1, 2, 3, 4 rows and
- y
- : 5, 10, 15, 20 tiles. How many tiles are in 12 rows?

Hint Find the amount for 1 first.

Show your work.

Answer _____ tiles

- 4.
- Gas Mileage.**
- A car goes the same distance on each gallon. A table shows
- x
- : 1, 2, 3 gallons and
- y
- : 30, 60, 90 miles. Write the equation. What does
- x
- stand for?

Show your work.

Equation _____

- 5.
- Ticket Booth.**
- Each ticket costs the same. Write an equation. Then find the cost of 14 tickets.

Hint Use one row to find the price of one.

Tickets (x)	3	6	9
Cost in \$ (y)	21	42	63

Equation; cost _____

- 6. Ravi's Rule.** A table shows x : 3, 5, 10 y : 27, 45, 90. Ravi wrote $y = x + 24$. Show that it does not fit every row. Then write the correct equation.
- Hint** Test the rule on the second row.

Show your work.

Equation _____

- 7. Bead Packs.** Every pack has the same number of beads. Write an equation. Tell what x and y stand for.
- Hint** Find the beads in one pack.

Packs (x)	4	7	10
Beads (y)	100	175	250

Equation _____

- 8. With a Constant.** A table shows x : 1, 2, 3 y : 9, 13, 17. $y \div x$ is not the same in every row. Write an equation. Then find y when $x = 10$.

Show your work.

Equation; y _____

FIND THE ERROR explain the mistake, then fix it

- 9. Callum's Work.** A table shows x : 2, 4, 6 and y : 14, 28, 42. Callum wrote the equation $x = 7y$. Explain his mistake and write the correct equation.

TEST PRACTICE choose the best answer

- 10.** For x : 1, 2, 3 and y : 5, 10, 15, the equation is –
- A** $y = x + 5$ **B** $y = 5x$ **C** $x = 5y$ **D** $y = x + 4$
- 11.** For x : 1, 2, 3 and y : 4, 7, 10, the equation is –
- A** $y = 4x$ **B** $y = x + 3$ **C** $y = 3x + 1$ **D** $y = 3x$

SELF-CHECK CODE find your answer · copy its letter into that problem's box · spell a math word

60 A	$y = 25x$ R	$y = 9x$ O	$y = 8x$ F	$y = 30x$ C	$x = 8y$ Q	16 Z	32 X	98 T

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #____
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